

DEVOPS IMPLEMENTATION FOR A COFFEE SHOP WEBSITE

Group 13

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PROJECT ON

DEVOPS IMPLEMENTATION FOR A COFFEE SHOP WEBSITE

Developed by

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DEVOPS IMPLEMENTATION FOR A COFFEE SHOP WEBSITE

Batch Code : 3FSD1

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Names of Developer :

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CERTIFICATE

This is to certify that this report titled “DevOps Implementation for a Coffee Shop Website” embodies the original work done by Namira Assalwa and Zulfarida. Project in partial fulfillment of their course requirement at NIIT.

Coordinator:

Rafif Agdyaputra, S.Kom

ACKNOWLEDGEMENT

First of all, we would like to express our deepest gratitude to God Almighty for His blessings and grace, which have enabled us to successfully complete our group project report entitled "DevOps Implementation for a Coffee Shop Website".

We would also like to express our sincere gratitude to our coordinator, Mr. Rafif Agdyaputra, S.Kom, for his invaluable guidance and input throughout the development of this project. His direction was fundamental in helping us fulfill our course requirements at CEP CCIT, Faculty of Engineering, Universitas Indonesia.

Our thanks also go to our committed team members, Namira Assalwa and Zulfarida from class 3FSD1, for their excellent cooperation and dedication. Each member played a crucial role in the success of this project, which was carried out from November 19th, 2025, to November 30th, 2025. The collaboration involved configuring Docker environments, managing Nginx web servers, and implementing GitHub Actions workflows.

Lastly, we appreciate the various learning resources regarding containerization and CI/CD that helped us overcome technical challenges during the project. We hope that this report can contribute positively to our learning journey and serve as a useful reference for future DevOps implementations.

SYSTEM ANALYSIS

System Summary:

The "DevOps Implementation for a Coffee Shop Website" project focuses on establishing a containerized infrastructure to support the web application. The core of the system is built using Docker technology, operating within an Ubuntu or Debian environment via WSL (Windows Subsystem for Linux) or Docker Desktop.

Technically, the architecture separates the application into distinct services: Nginx is utilized as the web server , while MySQL serves as the database management system. Configuration and orchestration are managed through essential files edited in VS Code, specifically the Dockerfile, dockerfile.nginx, docker-compose.yml, and default.conf, which define how the containers are built and networked. Furthermore, the system integrates GitHub Actions to handle the CI/CD workflow, ensuring automated and consistent deployment processes.

TECHNOLOGY USED

1. Operating system: Windows 11 Home Single Language
2. Docker: Desktop for Windows
3. Web server: localhost:1234
4. MySQL Database Server: mysql:3308
5. phpMyAdmin for Management Database: phpMyAdmin:8086
6. nginx system: php:8.2-fpm
7. WSL:
    a) Ubuntu
    b) Debian

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A) Dockerfile

Dockerfile is a text file (script) containing step-by-step instructions for building a Docker Image.

```
# ada di dalam nginx, secara fungsi sama tapi untuk membuat PHP-FPM
# untuk menjalankan kode PHP

# PHP-FPM untuk Nginx
FROM php:8.2-fpm

# Install MySQL extensions
RUN docker-php-ext-install mysqli pdo pdo_mysql

# Set working directory
WORKDIR /var/www/myphpapp

# Set proper permissions
RUN chown -R www-data:www-data /var/www/myphpapp

# Expose port 9000 untuk PHP-FPM
EXPOSE 9000
```

B) Dockerfile.nginx

Dockerfile.nginx is a naming variation of the standard Dockerfile, specifically dedicated to building an image for the Nginx web server service.

```
# Custom Nginx Dockerfile
FROM nginx
COPY ./default.conf /etc/nginx/conf.d/default.conf

# Expose port 80
EXPOSE 80

# Command untuk menjalankan nginx
CMD ["nginx", "-g", "daemon off;"]

# saat menjalankan nginx maka akan menjalankan kontainer (-g) secara otomatis akan running (daemon off)
```

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C) Docker-compose.yml

Docker-compose.yml is a YAML-formatted configuration file used by Docker Compose to define and run multi-container applications.

```yaml
# SETTING PORT
services:
  # Nginx Web Server (Custom Build) nampilin web pakai port 1234 = harus diubah tiap folder
  nginx:
    build:
      context: .
      dockerfile: Dockerfile.nginx
    ports:
      - "1234:80"
    volumes:
      - .:/var/www/myphpapp
    depends_on:
      - php
    networks:
      - app-network

  # PHP-FPM (untuk Nginx)
  php:
    build:
      context: .
      dockerfile: Dockerfile
    user: "1000:1000"
    working_dir: /var/www/myphpapp
    volumes:
      - .:/var/www/myphpapp
    depends_on:
      - mysql
    environment:
      - DB_HOST=mysql
      - DB_NAME=db_kantin
      - DB_USER=root
      - DB_PASSWORD=password
    networks:
      - app-network

  # MySQL Database Server = harus diubah tiap folder
  mysql:
```

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```
    image: mysql:8.0
    environment:
      MYSQL_ROOT_PASSWORD: password
      MYSQL_DATABASE: db_museum
      MYSQL_USER: appuser
      MYSQL_PASSWORD: password
    volumes:
      - mysql_data:/var/lib/mysql
    ports:
      - "3308:3306"
    networks:
      - app-network

  # phpMyAdmin untuk Database Management = harus diubah tiap folder
  phpmyadmin:
    image: phpmyadmin/phpmyadmin
    ports:
      - "8086:80"
    environment:
      - PMA_HOST=mysql
      - MYSQL_ROOT_PASSWORD=password
    depends_on:
      - mysql
    networks:
      - app-network

# Persistent Volume untuk MySQL Data
volumes:
  mysql_data:

# Custom Network untuk Container Communication
networks:
  app-network:
    driver: bridge
```

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D) Default.conf

Default.conf is the main configuration file for the "Server Block" on the Nginx web server.

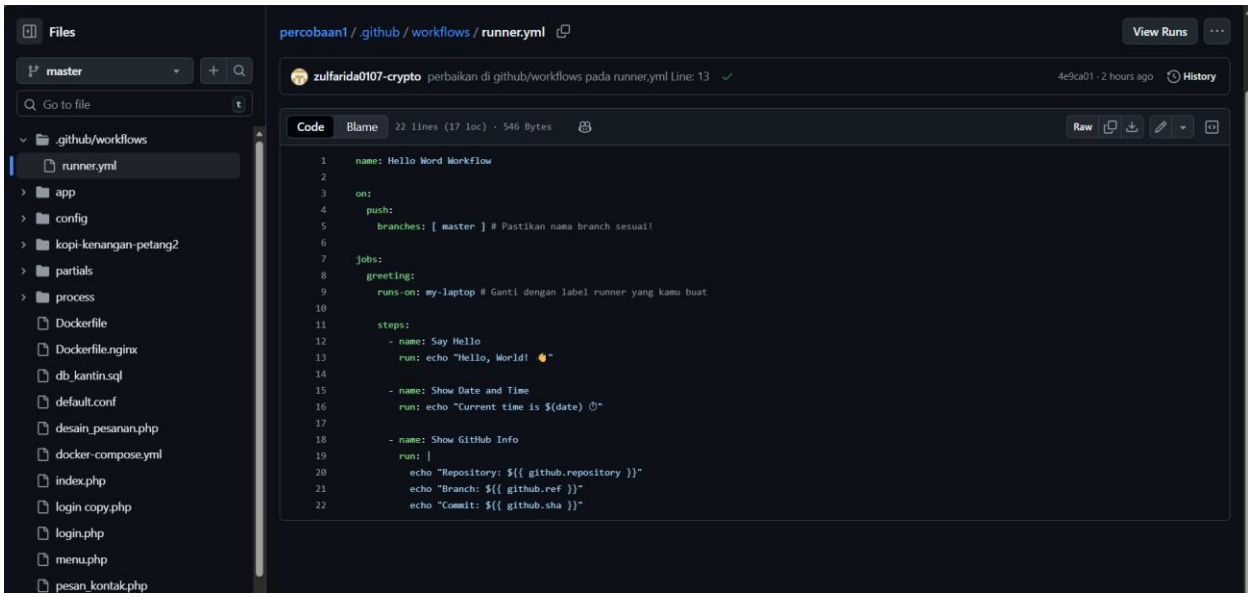
```
server {
 listen 80;
 root /var/www/myphpapp;
 index index.php index.html; #membaca di VS Code kalo ada file HTML akan dijalankan
 access_log /var/log/nginx/myphpapp_access.log;
 error_log /var/log/nginx/myphpapp_error.log;
 location / {
 try_files $uri $uri/ /index.php?$query_string;
 }
 location ~ \.php$ {
 fastcgi_pass php:9000;
 fastcgi_index index.php;
 fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
 include fastcgi_params;
 }
}

```

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E) Workflow from Github Actions

A GitHub Actions Workflow is a configurable automated process designed to execute software development lifecycles, such as building, testing, and deploying applications. This workflow is defined within a YAML file and is triggered automatically by specific events (such as code pushes) within the repository.



ACTIVITY LIST

A) Commit

A commit refers to the act of permanently saving changes made to the source code into the history of a version control system, such as Git. Technically, a commit serves as a checkpoint, capturing a snapshot of the state of all files at a specific point in time. Each commit possesses a unique identifier and a descriptive message (*commit message*), which facilitates developers in tracking the history of changes or reverting the application to a previous state in the event of errors.

